

Five layered Ensembled Deep Fully Connected Neural Network based Brain Stroke Prediction

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Abstract— Brain stroke infarction is triggered due to the blocking of blood passage in the brain. Researchers are experimenting the various genetic factors that influences the stroke. However, the exact identification of brain stroke remains a major challenge for the doctors. With this motivation, this paper aims to predict the existence of brain stroke by proposing Five Layered Ensemble Deep Fully Connected Neural Network (FLED) that uses the brain stroke database from the KAGGLE. The brain stroke sample has 10 characteristics and 4981 patient records that were encoded and treated with incomplete data. The exploratory data analysis have done to visualize the feature relations with the target. The dataset had applied to KNN, SVM, Decision tree, Random forest, XGB, ada boost, Gradient boost to analyze the performance of brain stroke prediction. Using a single input and output layer, the suggested Five Layered Ensemble Deep Fully Connected Neural Network was developed along with three hidden layers. The ensemble voting classifier had designed with five layered DFCNN, Ada Boost, Random Forest, Gradient Boosting and SVM to create an Ensemble Voting classifier and the performance was analyzed with all the classifiers. The proposed model Five Layered Ensemble Deep Fully Connected Neural Network have applied to the dataset to estimate the presence of brain stroke. Python was utilized for execution, and a server with a Geforce V100 Nvidia GPU was employed with a total sample size of 64 and 30 training iterations for analyzing the performance metrics. Experimental results shows that the proposed FLED model shows mean absolute error of 0.03745 and accuracy of 96.84% when compared with other classifier models.

Keywords— DFCNN, MAE, KNN, SVM, Voting classifier, accuracy.

I. INTRODUCTION

Brain stroke has developed into a serious illness that is currently fatal. Many people are losing their lives because of the difficulty and time-consuming nature of early brain stroke forecasting. In order to circumvent this, machine learning techniques determine the probability that a person is experiencing a brain stroke. A stroke is a medical disorder where there is inadequate blood flow to the brain, which results in cell death. Using properly trained data mining methods in fields like monitoring, healthcare, and information management, the application could reach the

public. The leading cause of adult mortality and disability globally is ischemic stroke. The rehabilitation choices made by doctors during the acute phase have a significant impact on each patient's diagnosis after a stroke. Hypertension represents one of the leading global causes of mortality and impairment, and its treatment and post-stroke care are expensive. The early detection of brain tumors and the correct cancer type would aid doctors in selecting the proper medications and in subsequent analysis based on how the individuals responded to the therapies.

II. LITERATURE REVIEW

The suggestion is to extract patients' ailments from case documents and use the information to teach the system. 507 outpatient report documents again from Sugam speciality Clinic were gathered during the phase of data gathering. The case sheets were then processed using maximum entropy and labelling approaches, and the suggested treatment collects the general and distinctive set of properties to categorize the disease [1]. Although thrombolysis in ischemic stroke patients is known to minimize disability and improve survival rates, some individuals still experience negative consequences. Therefore, being able to predict how patients with myocardial infarction would respond to systemic revascularization will be useful when making medical choices [2]. In this study, multiple linear regression (MLR) and two data mining approaches are used to create forecasting model, and the model performance is assessed using the Pearson coefficient of correlation between the features. Using hospital-based registry data connected to a judicial and administrative database, they evaluated these approaches in four distinct cohorts [3].

Numerous scores, including the ASTRAL, DRAGON, and THRIVE, have already been suggested over through the past five years as instruments to aid doctors in predicting the patient's prognostic value following a stroke. These evaluations use features that are accessible at the time and are based on rule-based classifiers. In this study, Predicting the operational outcomes of patients suffering from brain paralysis three months is a problem that is addressed by deep neural network approaches with the following hospitalization [4]. Using data that are easily accessible after treatment to the

emergency room, to create and to estimate operational prognosis in cerebral ischemia, verify a simple, arithmetic grading scale. In the origination sample of formerly independent persons with Information systems, regression was performed to explore the determinants of worse consequences. The estimated multidimensional model's covariate correlations were used to construct an rating system based on integers for each correlation coefficients, and the sum of the criteria weights was used to determine the final score. The model was externally tested in 2 separate cohorts and internally tested using 2-fold cross-validation method [5]. The recommendation made in the paper's conclusion is to use AUC rather than total accuracy when evaluating machine learning techniques on a single data point [6].

The findings from GBD 2016 represent the projections of the damage brought on by total stroke, ischemic stroke, and hemorrhagic stroke. Considering essential registry and linguistic pathology data as resources, an ensemble estimation method was used to determine cause-specific mortality. Bayesian meta-regression was used to create non-fatal estimates utilizing information from assessments, scientific literature, official statistics, and registry data [7]. Acute ischemic stroke can be accurately diagnosed by magnetic resonance imaging (MRI). The purpose of this study is to present an autonomous technique on enhancing learning argumentation framework using deformation key frames of MR images assisted reasoning system for determining the ischemic stroke. From the segmented lesions, morphological, anatomical and quantitative properties were collected to create a feature set that was further characterized using classification methods. The findings of the SVM classifier were inferior to suggested system, which accurately classifies the Radiofrequency cerebrovascular tumors with a 93.4% efficiency [8]. Machine learning models currently execute at levels that are tolerable, but they might be made to perform even better by incorporating optimization throughout the process of choosing features. The study presents a integrated sting ray exploring optimizer for image classification from malignancy MRI images. Deep convolutional system is employed to assess the optimal properties and potentially identify tumor cells. When compared to artificial neural networks and particle swarm optimization techniques, the modelling outperforms them in regards of detection and categorization accuracy [9]. This work seeks to consolidate and discuss the latest studies and advancements in frameworks, procedures, and solutions for data handling and management decision-making in diagnostics and prognostics of mechanical systems incorporating stroke. The authors also cover various methods for multiple sensor data fusion, as they are aware of using many sensors in condition monitoring [10]. In order to identify the type of cancerous tumors in the brain at their early stage, the paper proposes the capsule neural network in contrast to convolutional neural networks that can be practiced using a minimal group of datasets and is robust against spinning or interpolation conversions [11]. Gerontology and the intensity of a hemorrhage are quick, accurate markers of the likelihood of SICH and the treatment outcomes after Intravenous infusion therapy among numerous other clinical and imaging characteristics [12]. Numerous risk aspects have been linked in investigations to the start of brain hemorrhage in a patient. Data mining methods have been applied to forecast the stroke [13]. To increase the recognition rate, the stroke-affected regions of the brain was separated utilizing the fractional order Darwinian particle swarm

optimization technique, the partitioned volume was then analyzed using the anticipation methodology [14]. The evaluation period may be shortened, predictive performance may be increased, it may be easier to distinguish between different forms of ICH, and the risk of human mistake may be decreased using assisted or for brain stroke [15].

III. RESEARCH METHODOLOGY

The proposed FLED model fundamental research strategy is shown in Fig. 1. The framework of the recommended FLED model is illustrated in Figure 2.

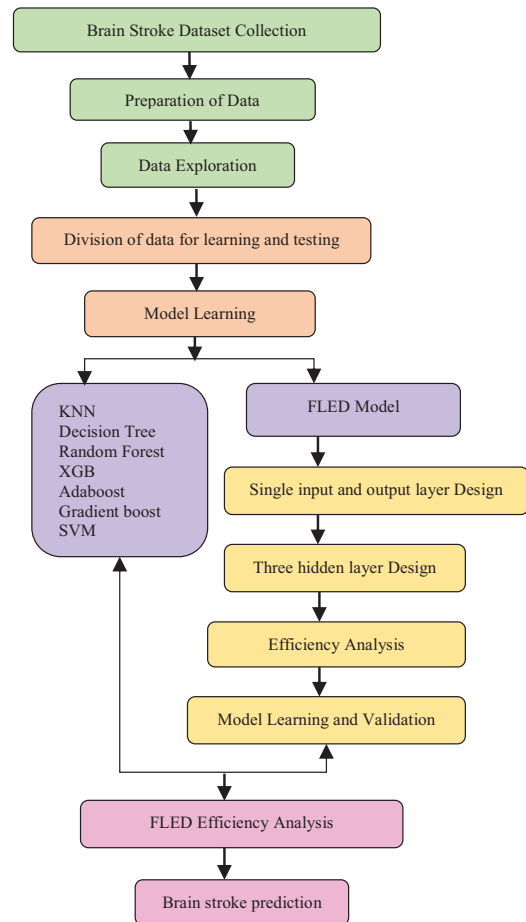


Fig. 1. Research Methodology

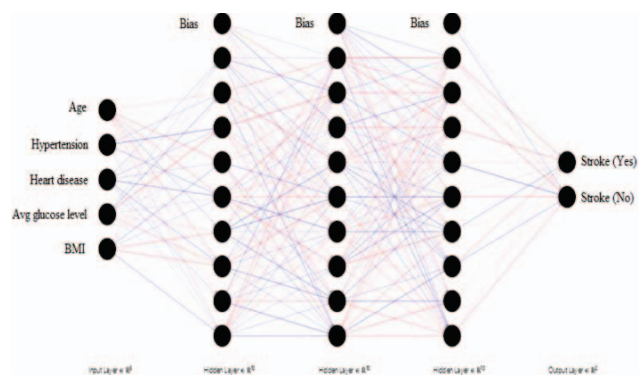


Fig. 2. FLED model

The primary focus of this work is to design the FLED model with one input and output layer, three hidden layers containing features like age, hypertension, heart disease, glucose level and BMI for predicting the existence of brain stroke. The brain stroke dataset contains 10 features with 4981 patient details and is used for brain stroke prediction is represented as in equation (1). The dataset is processed with encoding and missing values. Exploratory data analysis have done to visualize the correlation with the target.

$$X = \{[x1, x2, x3, \dots, x9], [Y]\} \quad (1)$$

Fig. 3. architecture

By calculating the average of each data for a given attribute, Equation (2) illustrates how to assess blank data after coding descriptive elements in a sample.

$$xR_{ij} = \frac{1}{n} \sum_{k=1}^m (xR_{ij})^k \quad (2)$$

Equation (3) demonstrates how to handle and examine the brain stroke statistics, including the computation of discrete data and categorical feature encoding.

$$X' = \frac{1}{x} \sum_{x=1}^9 X'_x \quad (3)$$

Where X' is the reprocessed collection excluding information loss and X is generated from the finalized information with the following independent attribute values: $x = 1, 2, \dots, 9$. The mean value of the processed variance for X'_x from the $x = 1, 2, \dots, 9$ is shown in (4).

$$W' = \frac{1}{x} \sum_{x=1}^9 X'_x = \frac{1}{x} \sum_{x=1}^9 var(X'_x) \quad (4)$$

The interpolated information W' is the one without any incomplete data. The following equation (5) is used to compute the component deviation within "BValue"

$$BValue = W' - \frac{1}{x-1} \sum_{x=1}^9 (X'_x - var(X'_x)) \quad (5)$$

The full resulting information set total correlation has been estimated and there are no null values by equation (6)

$$FinalDataset = W' + \left(\frac{x+1}{x}\right) \times BValue \quad (6)$$

The dataset is applied to KNN, SVM, Decision tree, Random forest, XGB, adaboost, Gradient boost to assess the efficiency of brain stroke forecasting. The proposed FLED is calculated with single input, three hidden layers and single output layer as shown in equation (7), (8), (9) and (10). The input layer is fed with the attributes Age, Hypertension, Heart disease, Avg glucose level and BMI. The three hidden layers contains the correlated values of five attributes from the input layer. The output layer contains the two classes as "Stroke and Absence of Stroke".

$$DFCNN = InLayer + 3 Hidden + OutLayer \quad (7)$$

$$InLayer_j^{[i]} = \sum_{i=1}^1 In_{j,i}^i \quad (8)$$

$$Hidden_j^{[i]} = \sum_{i=1}^3 Hid_{j,i}^i H_i^{i+1} \quad (9)$$

$$OutLayer_j^{[i]} = \sum_{i=1}^1 Out_{j,i}^i \quad (10)$$

The ensemble voting classifier is designed with five layered DFCNN, Ada Boost, Random Forest, Gradient Boosting and SVM to create an Ensemble Voting classifier and the performance is analyzed with all the classifiers as in equation (11). The training dataset is fitted with proposed model to analyze performance.

$$EVDFCNN - Y_{out} = Model \left\{ \begin{matrix} DFCNN(X), AB_{oo}(X), \\ RFor(X), SVM(X), GB_{oo}(X) \end{matrix} \right\} \quad (11)$$

Absolute mean error and Accurateness are the efficiency measurements used for investigation, as illustrated in equation (12), (13).

$$MAE = \frac{1}{m} \sum_{i=1}^m |y_{predicted} - y_{real}| \quad (12)$$

$$Accuracy = \frac{True Pos + True Neg}{True Pos + True Neg + False Pos + False Neg} \quad (13)$$

IV. IMPLEMENTATION SETUP AND RESULTS

The brain stroke dataset contains 10 attributes with 4981 patient details and is used for brain stroke forecasting. Python was utilized for execution, and a server with a Geforce V100 Nvidia GPU was employed with a total sample size of 64 and 30 training iterations for analyzing the performance metrics. The exploratory data analysis is shown in Fig. 3.

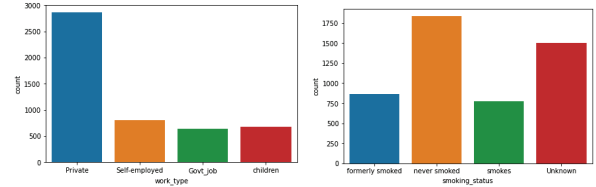


Fig. 4. Exploratory analysis of brain stroke dataset

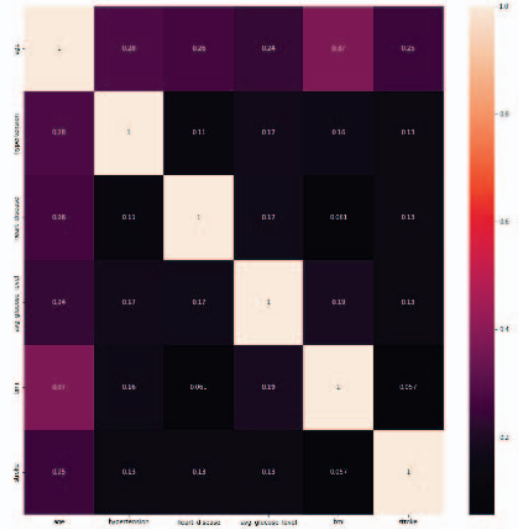


Fig. 5. Correlation of brain stroke dataset.

The dataset is applied to Five Layered Ensemble Deep Fully Connected Neural Network and the performance analysis is shown in Table I and Fig. 4 and Fig. 5. The 20% of the processed dataset is also validated with Five Layered Ensemble Deep Fully Connected Neural Network.

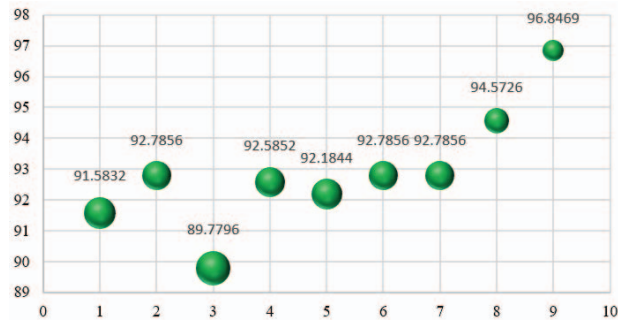


Fig. 6. Accuracy analysis of Five Layered Ensemble DFCNN

TABLE I. PERFORMANCE ANALYSIS OF FIVE LAYERED ENSEMBLE DEEP FULLY CONNECTED NEURAL NETWORK.

S.No	Model	Accuracy	Precision
1.	KNN	91.5832	0.0841
2.	SVM	92.7856	0.0721
3.	Dtree	89.7796	0.1022
4.	Rforest	92.5852	0.0741
5.	XGB	92.1844	0.0781
6.	Aboost	92.7856	0.0721
7.	Gboost	92.7856	0.0721
8.	5-DFCNN	94.5726	0.0524
9.	Ensemble 5-DFCNN	96.8469	0.03745

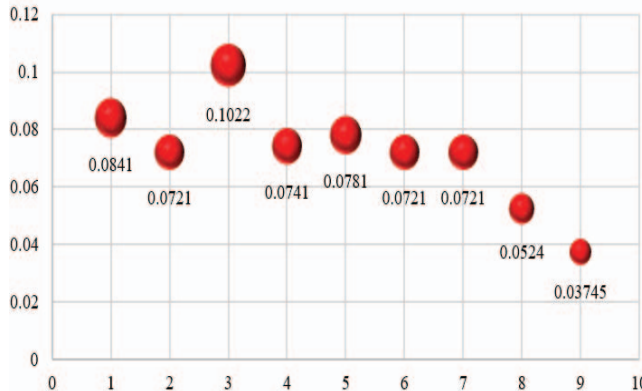


Fig. 7. Mean Absolute Error analysis of Five Layered Ensemble Deep Fully Connected Neural Network

V. CONCLUSION

This paper explores the novelty of projecting how well the various classifier models are performing when compared to the proposed Five Layered Ensemble Deep Fully Connected Neural Network model. An examination have been done with exploring the brain stroke dataset containing 10 features with 4981 patient details from Kaggle by analyzing the correlation of all features. The proposed FLED model is designed with single input, output layer and three hidden layers. The ensemble voting classifier is designed with five layered DFCNN, Ada Boost, Random Forest, Gradient Boosting and SVM to create an Ensemble Voting classifier and the performance is analyzed with all the classifiers. Python was

utilized for execution, and a server with a Geforce V100 Nvidia GPU was employed with a total sample size of 64 and 30 training iterations. Experiment results exhibits that proposed Five Layered Ensemble Deep Fully Connected Neural Network shows Mean absolute error of 0.03745 and accuracy of 96.84% when compared with other classifier models.

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